VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Major Project Phase - II Report

on

AI-Enabled Secure Mental Wellness Platform for Early Detection and Real-Time Emotional Support

Course Code: CSEP23706



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in

Computer Science and Engineering

Submitted by

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Department of Computer Science and Engineering

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Rajarajeshwari Nagar, Bengaluru - 560 098

2026-2027

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CERTIFICATE

Certified that the Major Project Phase-II work entitled **AI-Enabled Secure Mental Wellness Platform for Early Detection and Real-Time Emotional Support** has been successfully carried out at **Global Academy of Technology** by **Abhishek M H** bearing **1GA23CS004**, **Megesh H R** bearing **1GA23CS096**, **Nachikethana K B** bearing **1GA23CS103** and **Namith Nayak K** bearing **1GA23CS105** bonafide students of **Global Academy of Technology** in partial fulfillment of the requirements of final year degree in **Bachelor of Engineering in Computer Science and Engineering** during Academic Year **2026-27**. The Major Project Phase-II report has been approved as it satisfies the academic requirements in respect of project work for the said degree.

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DECLARATION

We, **Abhishek M H** bearing **1GA23CS004**, **Megesh H R** bearing **1GA23CS096**, **Nachikethana K B** bearing **1GA23CS103** and **Namith Nayak K** bearing **1GA23CS105** students of VII Semester B.E, Department of Computer Science and Engineering, Global Academy of Technology, Rajarajeshwari Nagar, Bengaluru, declare that the Major Project Work Phase-II entitled **AI-Enabled Secure Mental Wellness Platform for Early Detection and Real-Time Emotional Support** has been carried out by us and submitted in partial fulfillment of the requirements for the award of degree in Bachelor of Engineering in Computer Science and Engineering from Visvesvaraya Technological University, Belagavi during the Academic Year 2026 – 2027.

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ABSTRACT

Mental health problems such as stress, anxiety, depression, and emotional imbalance are increasingly affecting students and young adults in today's fast-paced environment. Due to social stigma, lack of awareness, and limited access to professional support, many individuals fail to receive timely assistance during critical situations. Existing digital mental health systems mainly focus on basic chatbot interaction or delayed consultation services and often lack continuous monitoring and early risk identification. To address these limitations, this project proposes an AI-Enabled Secure Mental Wellness Platform for Early Detection and Real-Time Emotional Support. The system combines standardized mental health assessment techniques such as PHQ-9 with Artificial Intelligence and Natural Language Processing methods to analyze emotional patterns and identify psychological distress at an early stage. The proposed platform provides real-time emotional support through AI-based conversational interaction and allows users to connect with mental health professionals through secure communication features when high-risk conditions are detected. The system also focuses on data privacy and secure user management to ensure confidentiality. This platform aims to improve accessibility to mental health support and encourage early intervention for better emotional well-being.

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Justification for Mapping SDG 3 to the Project

This project strongly aligns with **SDG 3 (Good Health and Well-Being)** by using Artificial Intelligence, Natural Language Processing, and standardized mental health assessment techniques to support early identification of emotional and psychological distress. The proposed platform focuses on conditions such as stress, anxiety, depression, and emotional imbalance, particularly among students and young adults. It provides AI-based emotional analysis, PHQ-9-based assessment, real-time conversational support, continuous monitoring, and professional consultation services. By enabling early detection and timely emotional assistance, the system aims to improve accessibility to mental wellness support and encourage early intervention. The secure handling of user information also supports the responsible delivery of digital mental health services.

SDG Contribution Statement:

- Promotes mental health and emotional well-being through AI-enabled support.
- Supports early identification of stress, anxiety, depression, and emotional distress.
- Provides real-time emotional assistance through an AI-powered conversational system.
- Enables connection with certified mental health professionals for high-risk users.
- Enables connection with certified mental health professionals for high-risk users.

Signature of the Student **Signature of the Project Guide**

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CHAPTER 1

INTRODUCTION

Walk through any college campus and you will find students who are tired, anxious or quietly falling behind. Exams, family expectations, social comparison and irregular routines all add up, and the number of students and young adults reporting stress, anxiety and low mood keeps climbing [1–4]. Many of them never ask for help. Some fear being judged, some do not realise that what they feel has a name, and some simply cannot find a counsellor who is free [5–8]. Conventional care also waits for the person to turn up at an appointment, which often happens only after things have already become serious [9, 10].

Advances in machine learning and natural language processing make it possible to notice distress earlier and to respond at once [11–13]. Our project, *AI-Enabled Secure Mental Wellness Platform for Early Detection and Real-Time Emotional Support*, which we call Healthly, is built around this idea. It pairs the PHQ-9 questionnaire with AI-based emotion analysis to flag early signs of distress [14–16], adds a chatbot and a peer community for everyday support, and lets a user speak to a professional over a live session when it is needed [17, 18]. Throughout, we paid attention to who can see a person's data, because a mental wellness tool that leaks records is worse than none at all [19, 20].

1.1 Problem Definition

Students and young adults are reporting more depression, anxiety and stress than before, yet most of them do not get help in time. Stigma, low awareness and a shortage of professionals all play a part. The digital tools available today usually do one thing: a chatbot that talks, or a booking page for a delayed appointment. They rarely watch a person's condition over time, rarely catch a worsening state early, and often handle personal data carelessly. What is missing is a single, secure platform that keeps track of emotional condition, spots high-risk situations early, and connects the person to support and to a professional without long delays.

1.2 Objectives

We set the following objectives for the work.

- Build a secure platform that can detect and monitor emotional distress at an early stage.
- Combine AI-based emotion analysis with standard questionnaires such as PHQ-9 to estimate depression, anxiety and stress.
- Offer immediate support through a conversational assistant and a peer community.
- Allow secure chat and video consultation with certified professionals for users at high risk.
- Make mental health support easier to reach so that people seek help sooner.

1.3 Scope of the Project Work

The platform is aimed at students and young adults who need support that is easy to reach and available at any hour. It covers standard screening questionnaires, AI-based emotion analysis, a chatbot and consultation with a professional.

Most of the analysis in this work is on text, where we use natural language processing and machine learning to find emotional patterns and risk levels. Voice, facial expressions, wearable data and medical reports are handled by separate models and merged later. Around these core features sit a community space, protected accounts and a video option for critical cases. We aimed for a service that can grow, that cuts the waiting time for help, and that protects privacy at each step.

1.4 Organization of the Report

Chapter 1 Introduction describes the background, the problem we address, our objectives and the scope of the work.

Chapter 2 Review of Literature surveys earlier research on AI for mental health, emotion analysis, assessment tools and chatbots, lists their weaknesses, and states the gaps that Healthly tries to fill.

Chapter 3 System Requirement Specification sets out what the system must do and what it needs in terms of hardware, software, models and data.

Chapter 4 System Design presents the architecture, data flow diagrams, use case and sequence diagrams, and the five modules.

Chapter 5 Implementation explains how the platform was built, the problems we met, and the algorithms behind the main workflows.

Chapter 6 Testing reports the unit, integration, system and functional tests we ran.

Chapter 7 Results and Discussion shows the working screens of the platform and discusses what they demonstrate.

Chapter 8 Conclusion and Future Scope summarises what was achieved, admits the limits of the current version and suggests how it could be extended.

CHAPTER 2

REVIEW OF LITERATURE

Interest in digital mental health has grown quickly because more people are struggling and there are too few professionals to help them. Researchers have tried emotion recognition, virtual counselling, AI-assisted diagnosis and personalised monitoring [1–4]. This chapter goes through twenty of these works, notes what each one does well and where it falls short, and then describes the survey we ran ourselves.

2.1 System Study

2.1.1 Overview of Existing Work

Each study below is summarised by its approach, its strengths, its limits, and what it means for Healthly.

Zhang, Huang and Liu [1] built AI-Edu-App, which puts schoolwork help and emotional wellbeing support in one application for K-12 students. Combining the two is a good idea because students already open the app for study. The cost is a more complicated system and more personal data to protect.

Soni and colleagues [2] describe Buddy, a chatbot that talks with people feeling stressed or anxious. It is available whenever the user wants it, which matters at night. Its weakness is that it often misses the real meaning of what a person says.

Yeasmin et al. [3] review how machine learning, chatbots, wearables and NLP can support behavioural health monitoring. They find that these tools widen access and allow continuous observation, but they also point to privacy, ethical questions and bias in the models.

In another paper [4], Yeasmin and co-authors look at AI in diagnosis and therapy. Early identification and tailored therapy are the gains, and the quality of the training data decides how far those gains go.

Zheng et al. [5] designed an emotion analysis and healing system for university students using deep learning and facial emotion recognition. It gives personalised support, though it needs heavy computation and struggles with subtle emotions.

The review in [6] surveys current trends in AI for mental healthcare. Its message is that AI helps with access and efficiency, yet it cannot supply the empathy of a human professional.

Boucher et al. [7] study chatbots used in digital mental health interventions, covering symptom monitoring and therapy support. Such chatbots scale well for people who cannot reach a clinician quickly, but the conversation is only as good as the data used to train them.

Vijayalakshmi, Sandhya and Deve [8] combine facial emotion recognition, a chatbot and a voice bot. Using several signals gives a fuller picture of the user, at the price of stronger hardware.

Ibitoye et al. [9] predict digital addiction from behaviour such as screen time and online activity and then suggest personalised support. The idea is useful for early warning, but it needs a large behavioural dataset.

Ma, Sun and Ma [10] propose PSG-Pair, a BERT-based model that matches questions and responses in wellness communities. It improves how relevant replies are, although noisy and unbalanced user data hurt it.

AlMakinah et al. [11] test large language models such as GPT-4, Mistral Large and Llama 3.1 for supportive conversation, judging quality and empathy. They raise worries about privacy, bias and trust, and suggest federated learning and review by professionals. For us the lesson is that a chatbot should work under professional oversight and never replace human care.

Zhang et al. [12] fine-tune wav2vec 2.0, with convolutional, attention and LSTM layers, to detect depression from speech. They show that transfer learning works even without a large labelled corpus. The result suggests speech as another signal for Healthly, though it must be validated on other datasets.

Vu et al. [13] compare Logistic Regression, Random Forest, Naive Bayes, SVM, XGBoost and LightGBM on NHANES data, using PHQ-9 scores. XGBoost did best, and SHAP showed which factors mattered. We take two points from this: PHQ-9 is a sound basis for prediction, and explanations make a model easier to trust. Results from one population may not carry over to another.

Oh et al. [14] introduce the AuD model, which uses DistilBERT to measure the intensity of depressive emotion and to classify symptoms in user utterances. This is finer than a simple depressed or not depressed label. It still depends on carefully annotated data.

Pinge et al. [15] review stress detection with wearables, from data collection to model training. Sensors can monitor a person between questionnaires and appointments, but their value depends on sensor quality and on the detection models. We treat wearable signals as supporting evidence and not

as a diagnosis.

Gomes et al. [16] survey wearable sensors for mental health and list measures such as heart rate, breathing, sleep and activity. These can show patterns linked with stress and anxiety. Sensor readings do not directly state an emotion, so they must be read alongside self-reports.

Srisankar and Lochanambal [17] compare LSTM-ANN and BiLSTM-ANN hybrids with plain LSTM and BiLSTM models. Sequence models capture order and context in text and time series, at the cost of more computation and some risk of overfitting. This supports our use of a BiLSTM for wearable data.

Sounack et al. [18] present BioClinical ModernBERT, an encoder for long biomedical and clinical text. A domain-specific model understands clinical language better than a general one, but it was not built for student emotional support, so we use it for medical reports only and would need to evaluate it before using it elsewhere.

Sanh et al. [19] describe DistilBERT, which keeps most of BERT's language understanding in a much smaller and faster model. That speed matters for a chatbot, since users will not wait. Smaller models can still make mistakes and need fine-tuning for the task.

Chen and Guestrin [20] describe XGBoost, a scalable gradient-boosted tree system that copes well with structured and sparse data. It suits the fusion stage of Healthly, where assessment scores and model outputs are combined, but it needs tuning and relevant mental health data.

Table 2.1: Literature Survey Table

Sl. No.	**Title of the Paper and Year**	**Methodology and Datasets Used**	**Merits**	**Limitations**
1	AI-Edu-App: An Integrated AI Platform for K-12 Learning Enhancement and Mental Health Support (2025) [1]	AI-based educational assistance and NLP-supported emotional monitoring	Combines learning support with emotional wellness	Complex implementation and privacy concerns

Sl. No.	Title of the Paper and Year	Methodology and Datasets Used	Merits	Limitations
2	Revolutionizing Mental Wellness: Buddy - An AI Powered Mental Health Chatbot (2025) [2]	NLP-based conversational chatbot for emotional interaction	Provides continuous emotional support	Limited contextual understanding
3	Artificial Intelligence for Mental Health Monitoring: A Solution for Digital Health (2026) [3]	AI-assisted monitoring and mental health prediction	Supports continuous emotional observation	Data privacy challenges
4	Artificial Intelligence in Mental Health: Leveraging Machine Learning for Diagnosis and Therapy (2025) [4]	Machine Learning-based emotional classification	Improves early mental health identification	Requires high-quality datasets
5	Design and Implementation of an Intelligent Emotional Analysis and Healing System (2025) [5]	Deep Learning-based emotion analysis system	Supports personalized emotional assistance	Prediction accuracy varies for complex emotions
6	Enhancing Mental Health with Artificial Intelligence: Current Trends and Future Prospects (2024) [6]	AI-driven emotional analysis and healthcare support	Improves accessibility of mental wellness systems	Limited human-like interaction

Sl. No.	Title of the Paper and Year	Methodology and Datasets Used	Merits	Limitations
7	Artificially Intelligent Chatbots in Digital Mental Health Interventions (2021) [7]	Chatbot-assisted digital mental healthcare	Provides scalable emotional support	Conversation quality depends on training data
8	AI-Powered Holistic Mental Health Monitoring Using Facial Emotion Recognition and Voicebot (2025) [8]	Facial emotion recognition and voice-based AI monitoring	Supports multimodal emotional analysis	High computational requirements
9	Predicting Digital Addiction Patterns with Machine Learning for Mental Health Support (2025) [9]	Behavioral prediction using Machine Learning models	Provides early addiction risk identification	Needs large-scale behavioral datasets
10	PSG-Pair: Psychological Snapshot Guided Model for Wellness Communities (2025) [10]	BERT-based emotional response pairing system	Improves contextual emotional response generation	Dataset imbalance affects performance

Sl. No.	Title of the Paper and Year	Methodology and Datasets Used	Merits	Limitations
11	Enhancing Mental Health Support through Human-AI Collaboration: Toward Secure and Empathetic AI-Enabled Chatbots (2024) [11]	Large Language Models (LLMs), human-AI collaboration, and privacy-aware chatbot design	Encourages empathetic support with human oversight	Privacy, bias, trust, and oversight challenges remain
12	Improving Speech Depression Detection Using Transfer Learning with wav2vec 2.0 in Low-Resource Environments (2024) [12]	Transfer learning using wav2vec 2.0 and deep learning for speech-based depression detection	Supports depression detection from speech with limited training data	Performance may depend on dataset quality and speaker diversity
13	Prediction of Depressive Disorder Using Machine Learning Approaches: Findings from the NHANES (2025) [13]	Machine learning models trained on NHANES data using PHQ-9-related information; XGBoost and SHAP for prediction and interpretation	Compares prediction methods and helps identify influential features	Generalization may be limited by dataset characteristics and population differences

Sl. No.	Title of the Paper and Year	Methodology and Datasets Used	Merits	Limitations
14	Are You Depressed? Analyze User Utterances to Detect Depressive Emotions Using DistilBERT (2023) [14]	DistilBERT-based analysis of user utterances for depressive emotion detection	Uses contextual language representations to identify depressive emotional patterns	Text-based predictions may miss non-verbal and situational information
15	Detection and Monitoring of Stress Using Wearables: A Systematic Review (2024) [15]	Systematic review of wearable-based stress detection and physiological monitoring methods	Highlights the potential of continuous stress monitoring	Sensor reliability and differences between studies affect comparison
16	A Survey on Wearable Sensors for Mental Health Monitoring (2023) [16]	Review of wearable sensors and physiological signals for mental health monitoring	Identifies opportunities for continuous and real-time monitoring	Wearability, data quality, privacy, and interpretation remain challenging
17	Deep Learning Models of LSTM-ANN and BiLSTM-ANN for Classification Accuracy (2024) [17]	LSTM-ANN and BiLSTM-ANN architectures for sequence modelling and classification	Explores deep learning methods for learning sequential patterns	Results depend on data quality, model configuration, and task-specific evaluation

Sl. No.	Title of the Paper and Year	Methodology and Datasets Used	Merits	Limitations
18	BioClinical ModernBERT: A State-of-the-Art Long-Context Encoder for Biomedical and Clinical NLP (2025) [18]	ModernBERT-based encoder designed for biomedical and clinical natural language processing	Supports processing of longer clinical texts and contextual information	Requires domain-specific validation and computational resources
19	DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter (2019) [19]	Knowledge distillation to create a smaller Transformer-based language model	Reduces model size and inference cost while retaining useful language understanding	May lose some capabilities compared with the larger BERT model
20	XGBoost: A Scalable Tree Boosting System (2016) [20]	Scalable gradient-boosted decision trees for classification and regression tasks	Provides efficient training and strong performance on structured data	Requires suitable feature engineering and parameter tuning

Besides the literature, we ran a survey of our own. A Google Form went to students at several colleges and asked about stress, anxiety, sleep, academic pressure, concentration and social life. About 1000 responses came back and were saved in a spreadsheet. A few questions were adapted from PHQ-9. We read the responses by hand and dropped the incomplete and contradictory ones before any analysis.

Figure 2.1 shows respondents by age group and gender. The 18–21 group is the largest, with roughly 90 male and 45 female respondents, and the counts drop sharply for older groups. The survey therefore mostly reflects undergraduates, the group Healthly is meant for.

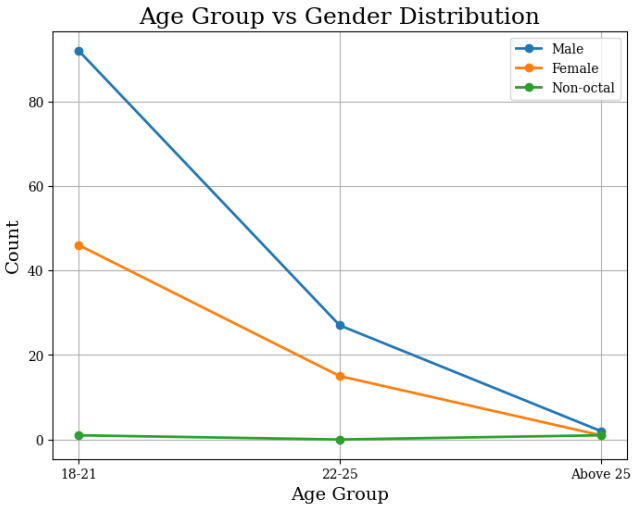


Figure 2.1: Age Group vs Gender Distribution

Peer pressure (Figure 2.2) is mostly reported as moderate, about 60 female and 55 male respondents, with fewer reporting low or high levels. Female counts are slightly higher in every category. Pressure from friends and competition for marks seems to be an ordinary part of student life, and it can wear down confidence over time, which is one reason early monitoring is useful.

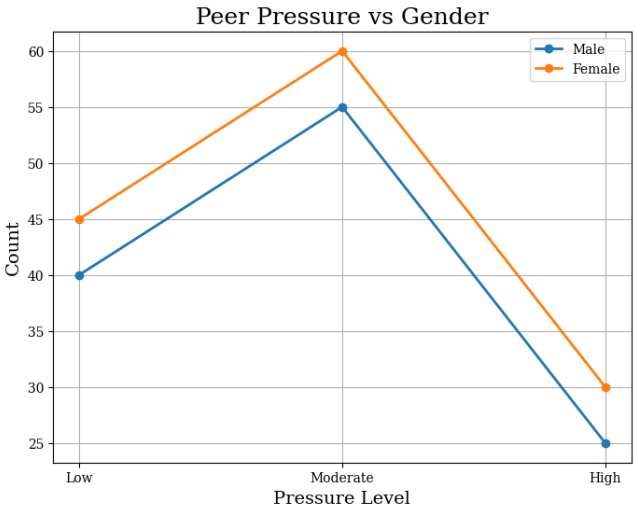


Figure 2.2: Peer Pressure vs Gender

Sleep is shown in Figure 2.3. Moderate sleep is the largest group at about 470, then good at about 320 and poor at about 210. A bad night makes it harder to concentrate the next day, and anxious people often cannot fall asleep, so the effect runs both ways. Healthly therefore records sleep beside mood and assessment scores and treats it as one clue among several, never as proof of a condition.

Finally, Figure 2.4 counts participation in physical activity. Walking leads with about 340 participants, then gym (210), running (180), sports (160) and yoga (95). Students who exercised regularly generally described themselves as calmer, but many said they could not keep a routine

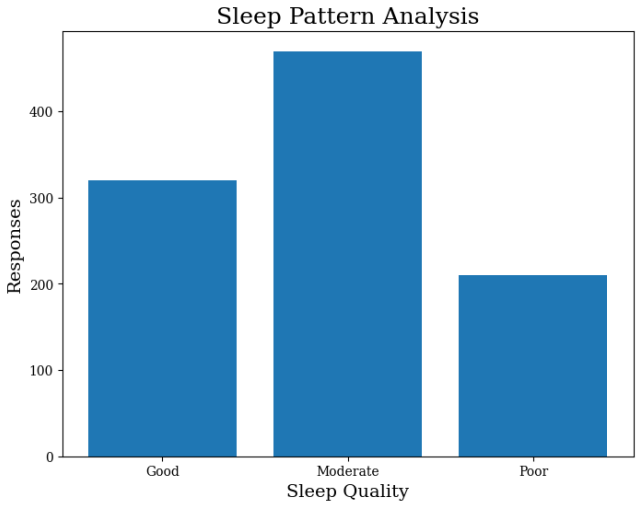


Figure 2.3: Sleep Pattern Analysis

because of workload. This is why Healthly reads activity data through Health Connect.

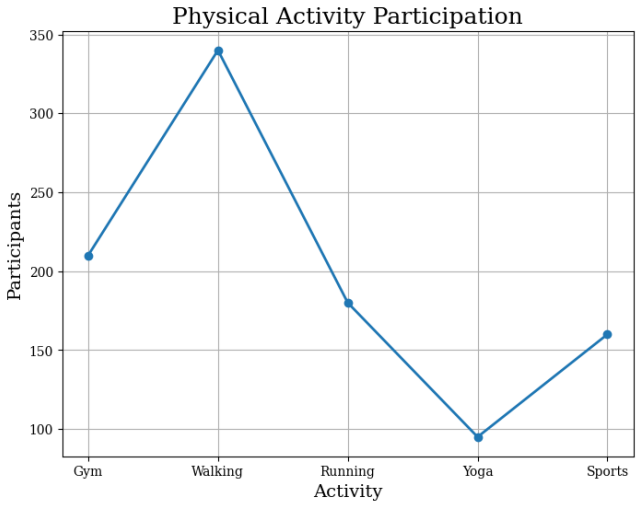


Figure 2.4: Physical Activity Participation

2.1.2 Research Gaps

Most of the systems reviewed do one job well, such as classifying emotion, chatting, or predicting a score. Very few bring monitoring, conversation, standard assessment and professional consultation together. Several also have weak privacy protection, limited understanding of context, heavy computation or slow real-time operation [3, 5, 7].

Second, few offer continuous tracking with early intervention for users at high risk. Third, many depend on general public datasets instead of responses from the students who will actually use the tool. Healthly is designed around these three gaps.

2.2 Proposed Work

We propose a secure platform that combines PHQ-9 screening with AI and NLP to estimate emotional state and detect risk early. It also provides a chatbot, emotional monitoring, a community space and consultation with professionals.

Text is handled with transformer models, namely DistilBERT for emotion and BioClinicalBERT for clinical records, while a BiLSTM reads wearable data, DeepFace and Wav2Vec2 handle face and voice, and an XGBoost classifier produces the final risk grade. The aim is support that is quick and continuous while personal information stays protected.

CHAPTER 3

SYSTEM REQUIREMENT SPECIFICATION

3.1 Functional Requirements

This section lists what Healthly must do from the user's point of view: sign-in, screening, analysis of text, voice, face and wearable data, risk grading and access to a counsellor.

Table 3.1: Functional Requirements of the Proposed System

Requirement	**Description**
Registration and Login	A user creates an account and signs in. Only a signed-in user can open personal records, so one person's data is never shown to another.
PHQ-9 Screening	The platform presents the nine PHQ-9 questions, adds up the answers and maps the total to a severity band.
Text Emotion Analysis	Chat messages and written answers go through an NLP model that labels the dominant emotion and reports its confidence.
Counsellor Access	A user can request a session with a certified professional and meet over secure chat or video.
Multimodal Analysis	PHQ-9 answers, chat text, wearable readings, facial cues and voice features are judged together, so the result never depends on one source alone.
Continuous Monitoring	Interactions and physiological or behavioural signals are watched for patterns that suggest a high-risk state so help can be offered early.

Requirement	Description
Risk Grading	The combined analysis is turned into a low, medium or high risk grade by the trained classifier.
Results Dashboard	Scores, detected emotions, the risk grade and suggested next steps appear on one readable screen.
Medical Report Analysis	Uploaded reports, prescriptions and clinical psychology notes are read by a clinical language model, which extracts diagnoses, medicines and risk details.
Wearable Data Processing	Heart rate, sleep, stress and activity trends from the user's phone or wearable are cleaned and analysed for signs of poor wellbeing.
Facial Emotion Recognition	Frames from a video feed are examined to estimate the emotion on the user's face.
Speech Emotion Recognition	Recorded voice is analysed for tone and stress-related features.

3.2 Non-Functional Requirements

Apart from features, the platform must meet quality targets covering speed, dependability, ease of use, protection of sensitive records and room to grow.

Table 3.2: Non-Functional Requirements of the Proposed System

Requirement	Description
Performance	Typical requests in the implemented modules should return in about 2–3 seconds under the light load of a prototype.

Requirement	**Description**
Reliability	The same input should give the same output, and processing failures should be rare and must not crash the application.
Usability	Screens should be uncluttered so that a student with no technical background can find the assessment, chatbot and consultation options unaided.
Security	Token-based login, role checks and encrypted traffic keep mental health records away from unauthorised people.
Privacy	Data is collected only after the user agrees, and personal, physiological and emotional records are stored so as to limit exposure.
Scalability	The design should let more models, monitoring features and healthcare services be added as usage grows.
Maintainability	Code is split into documented, independent modules so that a bug can be traced and fixed without touching unrelated parts.
Availability	The application should stay up during development, testing and demonstration, with only short interruptions.
Compatibility	The web application should open correctly on common operating systems and current browsers.
Extensibility	A new AI model, data source or consultation feature should plug into the existing structure without a redesign.

3.3 Hardware Requirements

The hardware below is what we used to build, run and test the platform, including capture of video, audio and wearable data.

Table 3.3: Hardware Requirement Specification

Component	**Specification**
Web Camera	Captures video for consultations and for facial emotion recognition.
Microphone	Captures voice for calls, spoken input and speech emotion recognition.
Smartphone-Based Sensor System	An Android phone running the Healthly companion APK collects activity, health and behavioural data through Android Health Connect and its built-in sensors. Recommended: quad-core processor, 4 GB RAM, 32 GB storage, Android 10 or later, Wi-Fi/Bluetooth, and motion and health sensors.
Controller	A controller module that reads the sensors and forwards readings to the processing system.
Network	A stable internet link for cloud services, API calls, live interaction and video sessions.

3.4 Software Requirements

The tools, languages, frameworks, models and services used to build Healthly are listed below.

Table 3.4: Software Requirement Specification

Component	**Specification**
Operating System	Windows 10/11, Linux or macOS for development, testing and deployment.
Frontend	React.js built with Vite; it renders the dashboard, chatbot, emotion monitoring views and analytics charts.
Backend Framework	FastAPI; it exposes the REST APIs, receives requests, calls the AI models and connects the modules.
Programming Languages	Python 3 for the backend, models and data handling; JavaScript for the client application.

3.4.1 Database Requirements

- **PostgreSQL:** Holds structured records such as user accounts and assessment results. The backend reaches it through SQLAlchemy.

3.4.2 AI Models and Frameworks

Each type of input goes to a model suited to it, and a fusion step merges their outputs.

- **DistilBERT:** Reads chat messages and PHQ-9 responses to detect emotion and context.
- **XGBoost:** Takes the fused features and assigns a low, medium or high risk grade.
- **BioClinicalBERT:** Extracts clinical information from uploaded reports, prescriptions and psychological records.
- **Bidirectional LSTM (BiLSTM):** Learns patterns in wearable time-series such as heart rate, sleep, stress and behaviour.
- **DeepFace CNN:** Estimates emotion from facial expressions in video frames.
- **Wav2Vec2:** Extracts emotional and stress-related cues from speech.
- **Attention-Based Fusion Engine:** Weights and merges the PHQ-9, text, sensor and video outputs into one representation.

3.4.3 Libraries and Frameworks

- **TensorFlow and PyTorch:** Build, train and run the deep learning models.
- **Scikit-learn:** Preprocessing, training utilities and evaluation metrics.
- **Pandas and NumPy:** Dataset handling, feature preparation and numerical work.
- **Transformers:** Provides the pretrained transformer models, DistilBERT in particular.
- **OpenCV:** Reads and processes images and video frames.
- **Matplotlib:** Plots data and model results.

- **LangChain:** Connects the chatbot to language-model components.
- **FastAPI and Flask:** Serve the backend APIs for requests, authentication and communication between the frontend and the AI modules.
- **React.js:** Builds the Healthly user interface.
- **Android Health Connect:** Supplies health and activity data from compatible Android apps and devices.

3.4.4 Development Tools and Platforms

- **Visual Studio Code:** Main editor for frontend and backend code.
- **GitHub:** Hosts the repository and tracks version history.
- **Overleaf:** Used to write this report in LaTeX.

3.4.5 Authentication and Security Requirements

Healthly stores personal details, credentials and what users write about their feelings, so security is applied at several levels instead of in one place.

- **JWT Authentication:** Every session carries a JSON Web Token that is verified before a protected service responds.
- **Fernet Encryption:** Sensitive fields are encrypted with a symmetric key, so stored values can be read back only by code that holds that key.
- **HTTPS:** Traffic between client and server is encrypted in transit.
- **Role-Based Access Control:** What a person can open depends on whether they are a user, therapist or administrator.
- **Consent-Based Data Handling:** Data is gathered and processed only with the user's permission.

3.4.6 Dataset Requirements

Our own survey supplied the training data. We circulated a Google Form among students and young adults and received roughly 1000 responses, which were kept as a spreadsheet.

The questions touched on stress, anxiety, sleep, general wellbeing, academic pressure and social contact. The data was used for preprocessing, PHQ-9 evaluation, behavioural analysis and model training. Before each model was trained, the responses were cleaned and reshaped to suit it.

CHAPTER 4

SYSTEM DESIGN

4.1 Design Overview

Healthly is designed to spot early signs of distress and to offer support at the moment it is needed. Rather than one large program, the platform is assembled from independent modules for login, screening, emotion analysis, chatbot conversation and counsellor access.

The frontend, backend, database and models exchange data through defined interfaces. Because modules do not depend on each other's internals, a model can be replaced or a feature added without breaking what already works, and user data stays protected throughout.

4.2 System Architecture

The architecture has seven layers, running from what the user sees down to the security controls. Each layer receives data from the one before and hands its result to the next.

Figure 4.1 shows the following layers:

- **User Layer:** Students and young adults use a web interface for assessments, chatbot support, wearable monitoring and counsellor sessions. Individuals, therapists and administrators have separate roles.
- **Assessment Layer:** Standard scales (PHQ-9, GAD-7, PSS-10 and WHO-5) give a first indication of depression, anxiety and stress.
- **Data Collection Layer:** Text, voice, facial expressions, wearable readings, medical reports and demographic details are gathered here to form a full picture of the user.
- **AI Analytics Engine:** Each modality has its own model: DistilBERT for text emotion, BioClinicalBERT for medical reports, BiLSTM for physiological signals, DeepFace CNN for faces and Wav2Vec2 for speech.
- **Fusion and Risk Estimation Layer:** An attention mechanism decides how much each modality should count, and the fused features go to an XGBoost classifier that returns a low, moderate or

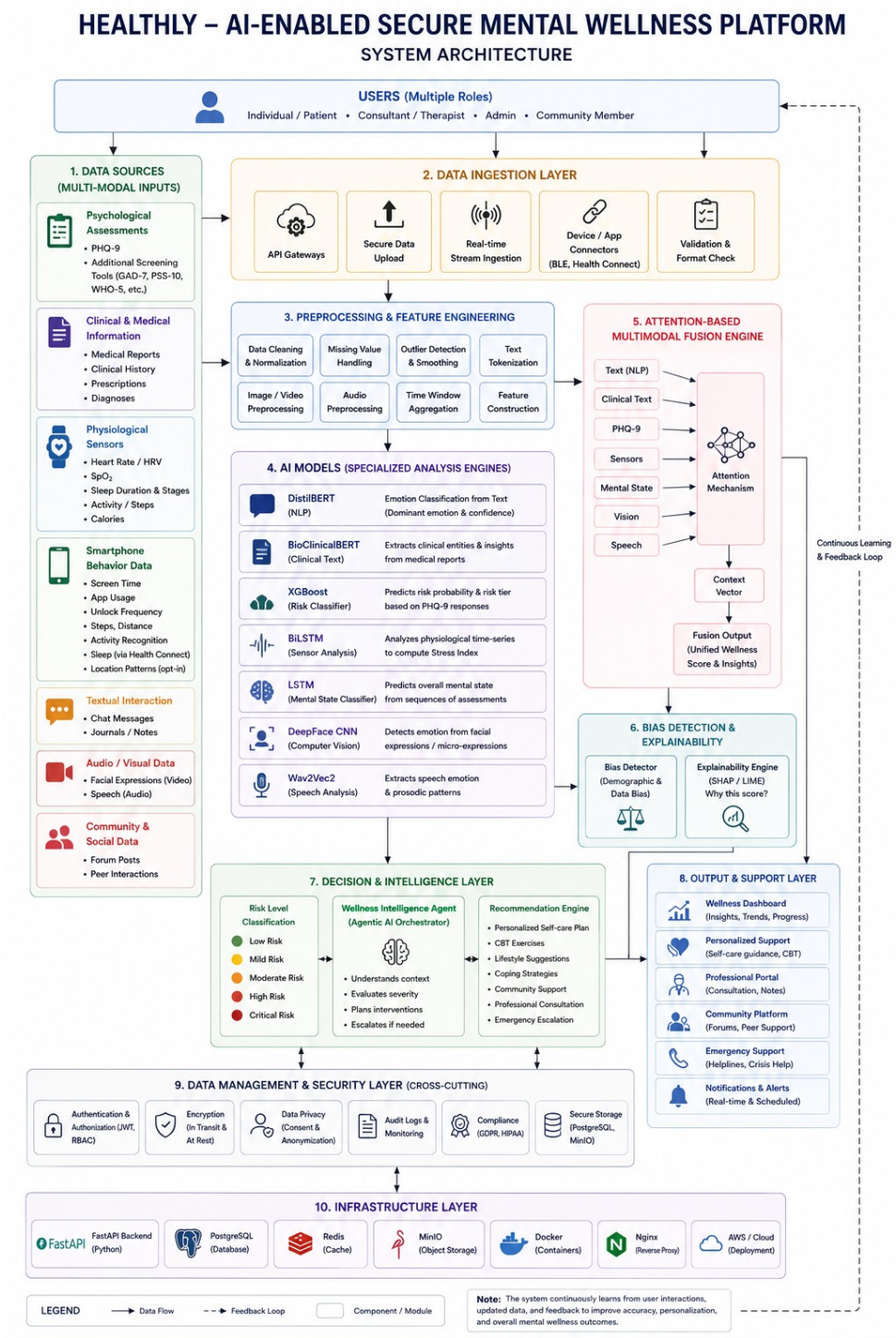


Figure 4.1: System Architecture

high risk grade.

- **Output and Support Layer:** According to the grade, the user receives insights, self-care suggestions and guidance. Counsellor sessions, community interaction and emergency help are offered where appropriate.
- **Security Layer:** JWT login, role-based access control, encryption, protected storage and privacy settings span all the other layers.

4.3 Data Flow Diagrams

4.3.1 Data Flow Diagram - Level 0

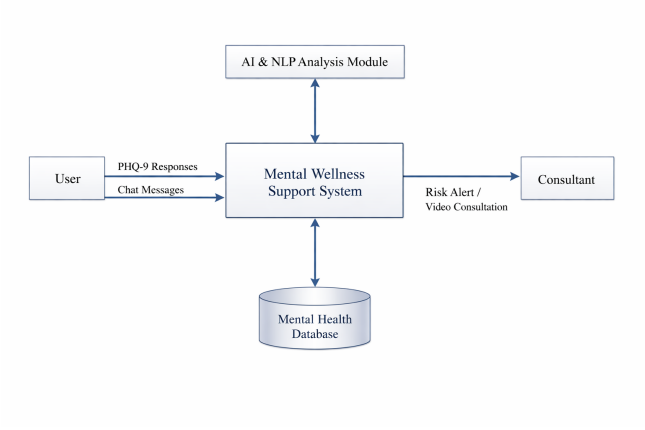


Figure 4.2: Data Flow Diagram - Level 0

Figure 4.2 treats the whole platform as one process. The user supplies PHQ-9 responses and chat messages, and the system draws on the AI and NLP module and the mental health database. It sends a risk alert or video consultation to the consultant when the situation calls for it.

Internal details are hidden at this level on purpose. For the user the visible result is a wellness report, chatbot replies, recommendations and access to professional help.

4.3.2 Data Flow Diagram - Level 1

Figure 4.3 opens the single process of Level 0 into its main modules: the PHQ-9 assessment, AI emotion analysis and the support decision engine. Their outputs feed AI chat support, community support and the consultant video call, and both the assessment and user data are written to the database.

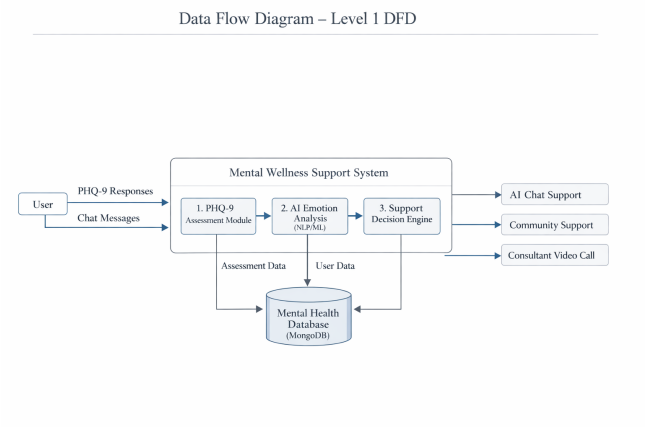


Figure 4.3: Data Flow Diagram - Level 1

In the full system the other inputs follow the same pattern. DistilBERT handles text and chat, BiLSTM handles wearable readings, DeepFace CNN and Wav2Vec2 handle face and voice, and BioClinicalBERT reads clinical documents. Their results go to the fusion engine, which produces the risk estimate used for insights and consultation advice.

4.3.3 Data Flow Diagram - Level 2

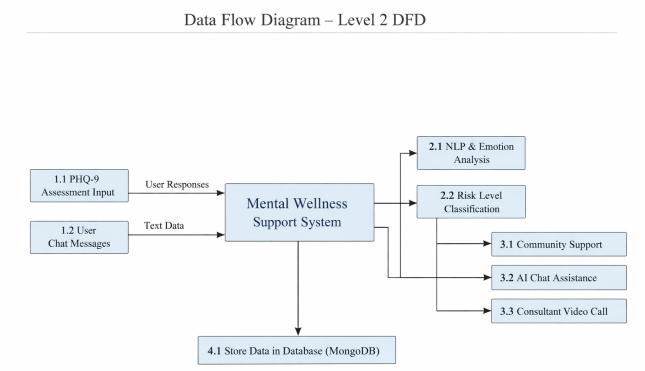


Figure 4.4: Data Flow Diagram - Level 2

Figure 4.4 zooms into analysis and support. Assessment input and chat text pass through NLP and emotion analysis and then risk level classification, after which the system triggers community support, AI chat assistance or a consultant video call. All results are saved to the database.

In practice a preprocessing step first removes incomplete, inconsistent or noisy entries. Attention-based fusion then judges which signals deserve more weight, and XGBoost turns the fused vector into a low, moderate or high class. Depending on that class the system issues recommendations,

chatbot support, consultation suggestions or emergency guidance, and keeps the outcome for later monitoring.

4.4 Use Case Diagram

Figure 4.5 shows how the actors relate to the services.

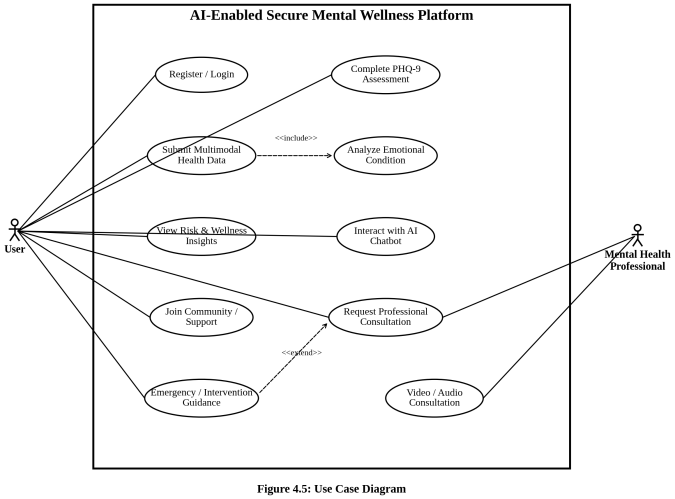


Figure 4.5: Use Case Diagram

The student is the main actor. After registering and logging in, the student can take the PHQ-9 test, send multimodal health data for analysis and see risk and wellness insights.

The chatbot is available whenever conversation is wanted, and the community feature allows supportive exchanges with peers. When professional help is needed, the student can book a video or audio session with a mental health professional, and an emergency guidance feature covers urgent situations. Every service is reached from the same application.

4.5 Sequence Diagram

Figure 4.6 follows one complete session in time order.

The session starts when the user logs in. The backend checks the credentials against the database and, if they match, opens the dashboard.

The user then submits PHQ-9 answers, which the backend stores before showing the assessment status. Text or audio given to the support assistant is stored and passed for emotion analysis.

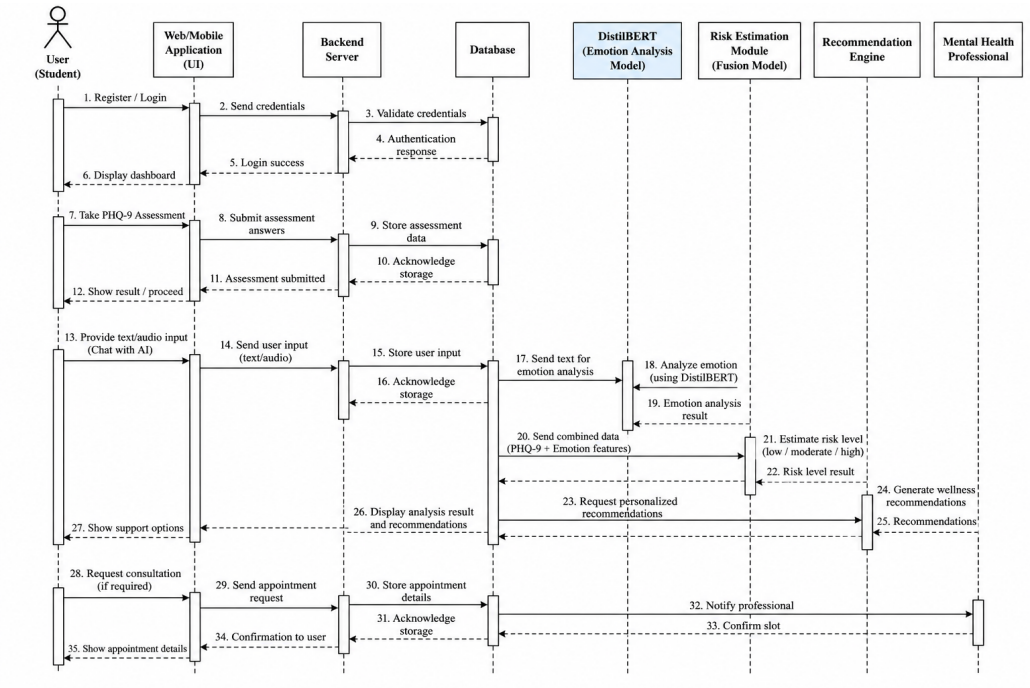


Figure 4.6: Sequence Diagram

DistilBERT returns an emotion result, and this, together with the assessment data, goes to the risk estimation module, which assigns a low, moderate or high level.

The recommendation engine uses that level to prepare personalised suggestions, which come back to the interface with support options. If the user asks for a consultation, the backend saves the appointment and notifies the professional, whose confirmation of a slot is sent back to the user.

4.6 Modules of the Project

The project is divided into five modules:

1. Module 1: User Authentication Module
2. Module 2: Mental Health Assessment Module
3. Module 3: Emotion Analysis / AI Analytics Module
4. Module 4: AI Chatbot & Emotional Support Module
5. Module 5: Professional Consultation Module

4.6.1 Module 1: User Authentication Module

Functionality: Registers users, logs them in and restricts each person to their own data and permitted services.

Input: Registration details, email or username, password and role.

Output: A verified session and access to the features allowed for that role.

Algorithm used: JWT-based authentication with Role-Based Access Control (RBAC).

4.6.2 Module 2: Mental Health Assessment Module

Functionality: Screens the user with standard questionnaires such as PHQ-9 to detect early signs of depression, anxiety and stress.

Input: Answers to PHQ-9 and other screening questions.

Output: A total score and the matching severity or risk indication.

Algorithm used: PHQ-9 score summation with rule-based severity bands.

4.6.3 Module 3: Emotion Analysis / AI Analytics Module

Functionality: Studies text, speech, facial expressions, physiological signals and clinical records to find emotional patterns, stress indicators and behavioural change.

Input: Chat text, voice samples, facial expressions, wearable readings and medical reports.

Output: Detected emotions and the feature vectors needed for risk estimation.

Algorithm used: DistilBERT (text), BiLSTM (wearable data), DeepFace CNN (faces), Wav2Vec2 (speech) and BioClinicalBERT (clinical text).

4.6.4 Module 4: AI Chatbot & Emotional Support Module

Functionality: Holds a live conversation with the user, offers wellness guidance and suggests further help when it notices unusual emotional signals.

Input: Chat messages, emotion results, assessment results and earlier conversations.

Output: Context-aware replies, wellness suggestions and, when required, advice to seek professional or emergency help.

Algorithm used: Transformer-based NLP, namely DistilBERT, to interpret the emotional context of each message.

4.6.5 Module 5: Professional Consultation Module

Functionality: Links users with certified mental health professionals, especially when the risk level is high, through video or audio sessions.

Input: Assessment results, risk level, consultation request and the professional's availability.

Output: A consultation recommendation, a scheduled slot and the video or audio session itself.

Algorithm used: XGBoost risk classification into low, moderate and high; high-risk users are advised to consult a professional.

CHAPTER 5

IMPLEMENTATION AND RESULTS

This chapter explains how we turned the design of Chapter 4 into a running application. The work covered the user interface, backend services, database, model integration and testing, with debugging going on throughout.

5.1 Steps for Implementation

We built the system in stages so that each part could be connected to the others one at a time.

- **Setting up the environment:**
  React.js, Vite, Python, FastAPI, PostgreSQL, Visual Studio Code and Overleaf were installed and configured.
- **Frontend:**
  With React.js and Vite we built the login, assessment, chatbot and consultation screens.
- **Backend:**
  FastAPI endpoints were written for authentication, assessment scoring, emotion analysis, chatbot messages and database access.
- **Dataset collection and preprocessing:**
  Survey responses from students and young adults were gathered through Google Forms, stored as a spreadsheet and cleaned for emotion analysis and model training.
- **Model integration:**
  DistilBERT, BioClinicalBERT, BiLSTM, DeepFace, Wav2Vec2 and XGBoost were connected to the backend for emotion detection and risk classification.
- **Testing and debugging:**
  Each module was tested alone and then together, and functional, logic and integration faults were corrected.

- **Result verification:**

  The complete system was run with sample users, PHQ-9 answers and chatbot conversations to check that the results were sensible.

5.2 Implementation Issues

Getting React.js, FastAPI, the NLP models and the real-time communication components to work as one system was the hardest part. Mismatched data formats and timing between modules caused errors that took repeated debugging and testing to track down.

We also had to handle user data, emotion results and chatbot replies quickly without weakening security. To keep later changes manageable, we kept the code modular and readable.

Our aims during implementation were:

- Keep memory use low while the models run.
- Keep the source code readable and easy to maintain.
- Improve the accuracy of emotion analysis results.
- Shorten the response time seen by users.
- Protect user information and assessment data.
- Make it simple to add more models and support modules.

5.3 Algorithms

5.3.1 Multimodal Mental Wellness Assessment and Risk Prediction

1. The user logs in securely.
2. The system collects inputs from the available sources:
   - PHQ-9 responses
   - Chat conversations
   - Wearable sensor readings

- Medical reports
- Video and speech

3. PHQ-9 answers are converted into descriptive text so they can be interpreted in context.
4. Incomplete, inconsistent or noisy data is removed.
5. DistilBERT and BioClinicalBERT process the conversational and clinical text.
6. A BiLSTM analyses physiological data such as heart rate, sleep and activity.
7. DeepFace CNN recognises facial emotion and Wav2Vec2 analyses speech emotion.
8. The Attention-Based Fusion Engine merges the outputs of all modalities.
9. The XGBoost classifier assigns a low, moderate or high risk level to the fused result.
10. A wellness score, recommendations and support suggestions are generated.
11. If the condition is critical, the user is directed to professional consultation and emergency services.
12. Results are stored securely for future monitoring.
13. The process ends.

5.3.2 AI Chatbot and Real-Time Emotional Support

1. The user starts a chatbot session.
2. The system receives the user's messages and any other conversational input.
3. The text is cleaned using standard NLP preprocessing.
4. DistilBERT analyses the emotion and sentiment of the conversation.
5. The system checks for stress, anxiety, sadness, hopelessness or emotional imbalance.
6. The conversational engine writes a supportive reply that matches the context.

7. The conversation is monitored continuously for high-risk behaviour.
8. If something abnormal is found, an alert is raised and professional consultation is suggested.
9. The chat history is saved securely so trends can be studied later.
10. The loop continues until the user leaves the session, and then the process ends.

5.3.3 Wearable Sensor and Physiological Monitoring

1. The Healthly companion app on the Android phone reads data through Android Health Connect and the phone's sensors.
2. The data collected includes:
    - Heart rate
    - Sleep patterns
    - Physical activity
    - Stress indicators
3. The readings are sent to the FastAPI backend.
4. They are preprocessed and normalised.
5. A BiLSTM model studies the temporal behaviour patterns.
6. Stress level and emotional condition are estimated from the continuous data.
7. The result is passed to the fusion engine for multimodal evaluation.
8. Wellness insights and physiological trends appear on the dashboard.
9. Sensor history is stored for ongoing monitoring.
10. The process ends.

CHAPTER 6

TESTING

Testing checked that the main modules of Healthly give correct results and work correctly together. We looked at single functions, the links between modules, whole user workflows and the screens a user actually sees, using both valid and invalid inputs.

6.1 Test Environment

Tests were run locally. The frontend (React.js with Vite) ran on the Vite development server and the backend (Python, FastAPI) ran on Uvicorn, with the application opened in a modern browser. SQLAlchemy connected the backend to the project database, which holds users, assessments, chats, sensor data, health reports, forum posts and consultations.

The AI components used Transformers, PyTorch, XGBoost, Scikit-learn, OpenCV, DeepFace and Cryptography. Separate Python test scripts verified PHQ-9 analysis, emotion classification, multimodal processing and the machine learning models. Security features, namely JWT authentication, password hashing and Fernet encryption and decryption, were checked as part of this testing.

6.2 Unit Testing

Unit tests exercised individual functions and services on their own, before they were connected to other modules.

Table 6.1: Unit Testing

Steps	**Test Data**	**Expected Results**	**Observed Results**	**Remarks**
Step 1	PHQ-9 answers, each between 0 and 3	Total score is computed and a severity level assigned.	Score and severity level came out correctly.	Pass
Step 2	Text: "I feel very sad and hopeless about everything."	An emotion label is returned with a confidence value.	The classifier returned a label with its confidence score.	Pass
Step 3	Plain text and sensitive data given to the encryption service	Data is encrypted and then decrypted with the configured Fernet key.	Encryption and decryption routines protected and recovered the stored data.	Pass

6.3 Integration Testing

Integration tests checked that data moves correctly between the frontend, backend, database, AI services and supporting modules.

Table 6.2: Integration Testing

Steps	**Test Data**	**Expected Results**	**Observed Results**	**Remarks**
Step 1	Valid registration details	Frontend sends the data and a new user is created in the database.	The authentication API accepted the data and saved the user.	Pass

Steps	Test Data	Expected Results	Observed Results	Remarks
Step 2	Registered email and correct password	Backend verifies credentials and returns a token.	Login succeeded and a JWT access token was issued.	Pass
Step 3	Valid PHQ-9 answers	Score and risk level are calculated, emotion and risk analysis run, and the assessment is stored.	The PHQ-9 endpoint scored, analysed, classified, produced recommendations and saved the record.	Pass
Step 4	PHQ-9 results plus a chat message	Chatbot uses the assessment context in its reply.	The chat service fetched the latest PHQ-9 result and used it with the message.	Pass
Step 5	PHQ-9, medical, sensor, chat and video data	All available modalities are processed into one risk assessment.	The multimodal workflow combined the sources through attention-based fusion.	Pass
Step 6	Heart rate, HRV, sleep duration and activity	Stress measures and physiological risk are calculated.	The wearable analysis service returned stress and risk indicators.	Pass

Steps	Test Data	Expected Results	Observed Results	Remarks
Step 7	Medical text with diagnoses, medications and clinical details	Relevant clinical information and risk indicators are identified.	The medical analysis service extracted entities, diagnoses, medications and risk information.	Pass
Step 8	Chat, facial-expression and speech data	Emotional analysis is produced from all three sources.	The services returned emotion, sentiment, facial and speech indicators.	Pass
Step 9	Consultation request and professional availability	Request is saved and user and professional can communicate.	The consultation and video-call modules handled the appointment and communication flow.	Pass

6.4 System Testing

System testing ran the whole application end to end, covering login, assessment, AI analysis, monitoring, recommendations and professional support.

Table 6.3: System Testing

Steps	Test Data	Expected Results	Observed Results	Remarks
Step 1	New user registers and logs in	Account is created and protected pages open.	Registration, login, JWT authorisation and page protection all worked.	Pass

Steps	Test Data	Expected Results	Observed Results	Remarks
Step 2	Full PHQ-9 with valid answers	Score and classification are displayed.	Score, risk, emotion analysis and recommendations were returned.	Pass
Step 3	Chat messages seeking support	Chatbot analyses input, replies and records the conversation.	A reply was generated and the conversation saved in the database.	Pass
Step 4	Multimodal data from the available modules	Overall wellness and risk assessment is produced.	Data passed through the fusion and safety analysis layers.	Pass
Step 5	Professional consultation request	User can request help and use the consultation workflow.	Appointment and professional communication features worked.	Pass

6.5 Functional Testing

Functional testing confirmed that each feature does what a user expects, judged from the input given and the behaviour observed.

Table 6.4: Functional Testing

Steps	Test Data	Expected Results	Observed Results	Remarks
Step 1	Registration and login with valid credentials	User is authenticated and reaches the correct dashboard.	Valid credentials gave authorised access.	Pass

Steps	Test Data	Expected Results	Observed Results	Remarks
Step 2	PHQ-9 with valid answers	Total score and assessment result are shown.	Score, severity, emotion data and recommended action were displayed.	Pass
Step 3	Message sent to the chatbot	A conversational reply with emotion information is returned.	The message was analysed and the reply and analysis stored.	Pass
Step 4	Sensor, medical, chat or video data	Wellness, risk and recommendation details are displayed.	The analysis and multimodal modules returned results.	Pass
Step 5	Consultation request	User can start the consultation and see appointment details.	Consultation requests and video/audio communication worked.	Pass

CHAPTER 7

RESULTS AND DISCUSSION

This chapter walks through the working Healthly application. Each screenshot shows a real screen from the build, covering assessment, AI analysis, sensor data, consultation and administration in one platform.

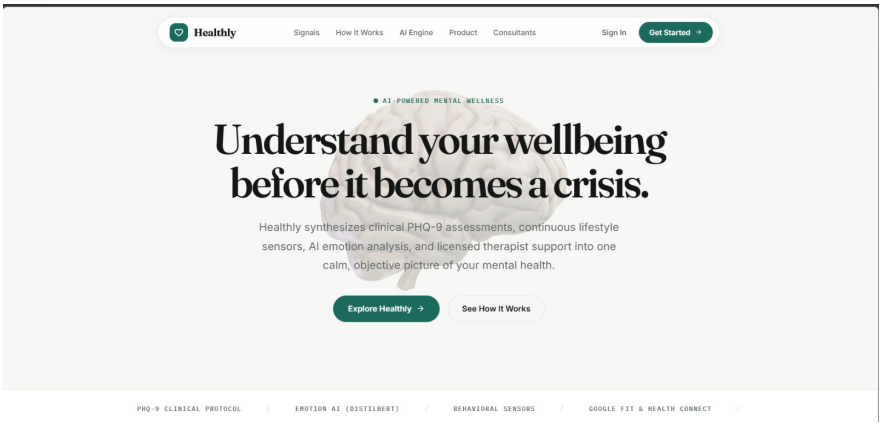


Figure 7.1: Healthly Platform Landing Page

The landing page is the first thing a visitor sees. Its headline asks people to understand their wellbeing before it becomes a crisis, and it lists PHQ-9, emotion AI, behavioural sensors and Health Connect as the main features.

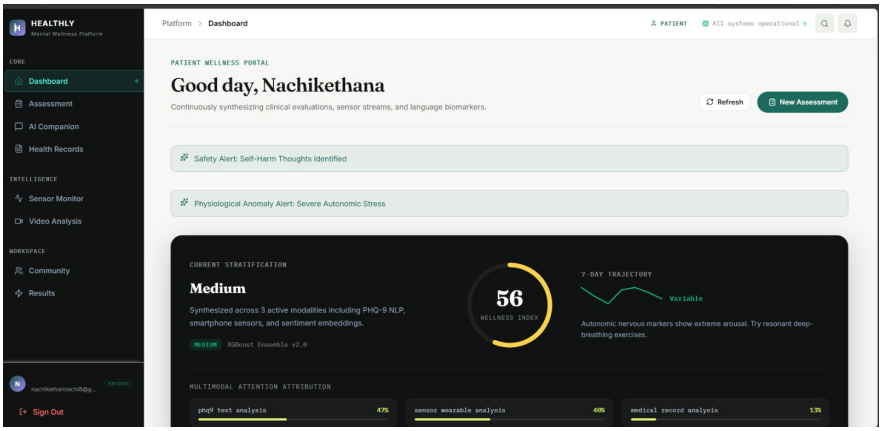


Figure 7.2: Patient Wellness Dashboard

After login, the patient dashboard shows a wellness score (56 in this sample, graded Medium), a seven-day trajectory, safety and physiological alerts, and how much weight each data source received in the fused result.

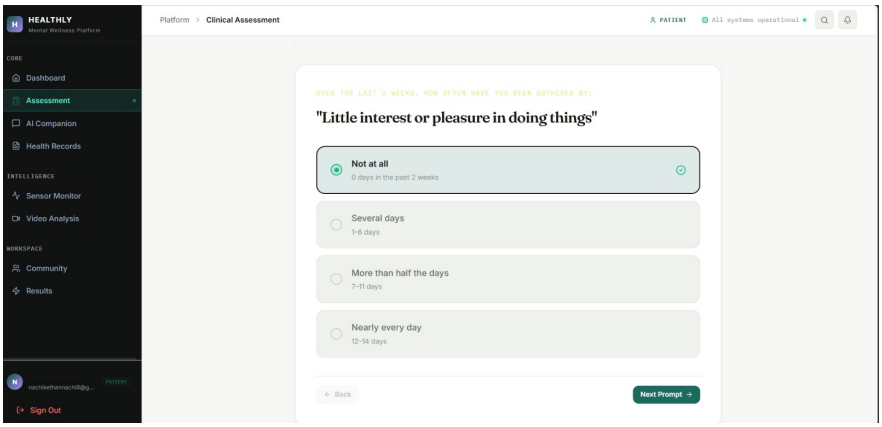


Figure 7.3: PHQ-9 Clinical Assessment

The assessment screen shows one PHQ-9 question at a time, for example “Little interest or pleasure in doing things”, with four options from Not at all to Nearly every day.

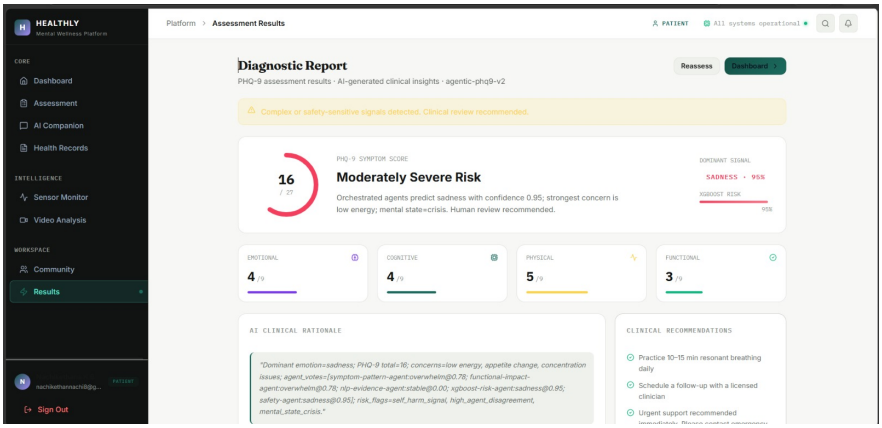


Figure 7.4: AI-Based Assessment Results

The results page reports a PHQ-9 score of 16, labelled moderately severe risk, with sadness as the dominant emotion. It also gives a short AI rationale and clinical recommendations such as breathing exercises and a follow-up with a licensed clinician.

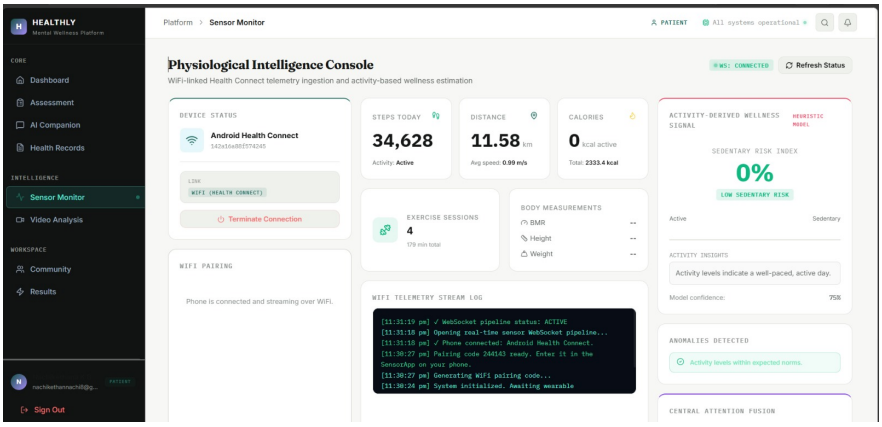


Figure 7.5: Physiological Sensor Monitoring

The sensor console shows live Health Connect data from a paired Android phone, including 34,628 steps, 11.58 km distance and four exercise sessions, together with a sedentary-risk estimate and a connection log.

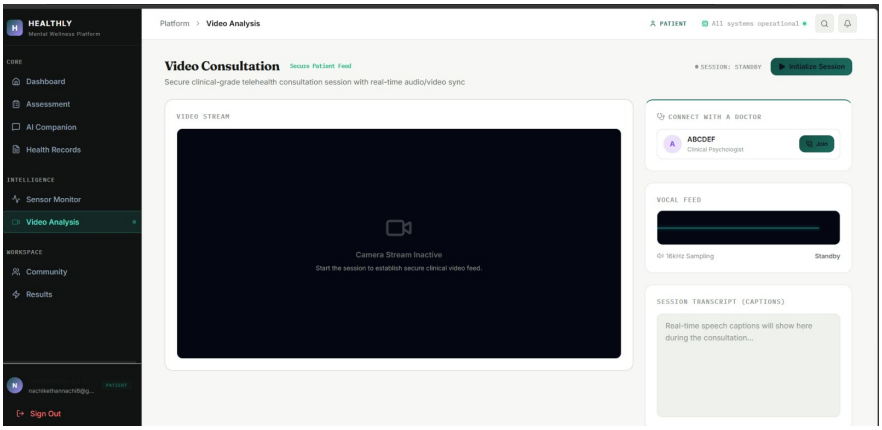


Figure 7.6: Secure Video Consultation Interface

The consultation screen contains a video area, a list of professionals to connect with, a voice feed and a space for live captions during the session.

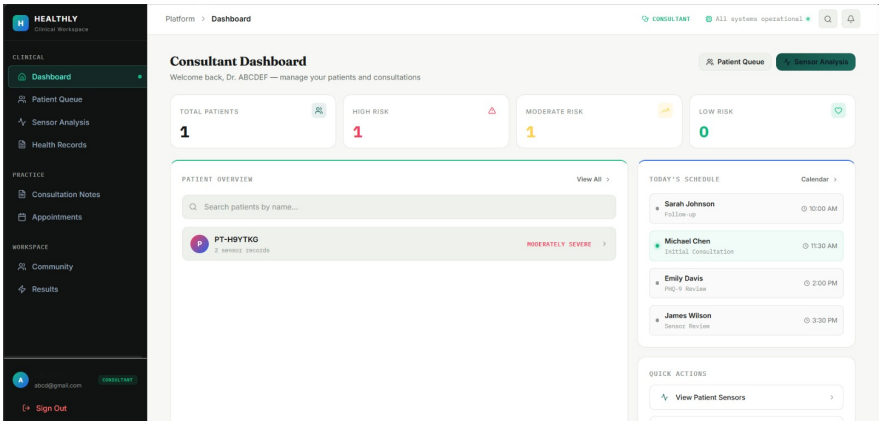


Figure 7.7: Mental Health Professional Dashboard

The consultant dashboard counts patients by risk level, lists the patient queue and the day's appointments, and links to each patient's sensor data and records.

The administrator view gives totals of users, patients, consultants and assessments (14, 4, 8 and 15 in this sample), a user management table, and system health indicators such as API latency and uptime.

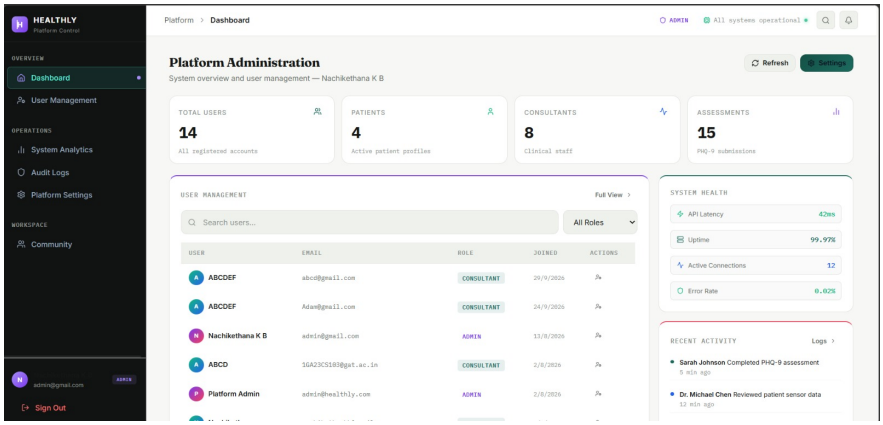


Figure 7.8: Administrator Dashboard

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

We set out to build one platform where a student can be screened, listened to, monitored and, if needed, connected to a professional. Healthly now does this. PHQ-9 answers, chat text, wearable readings, medical reports, voice and facial expressions are each handled by a suitable model, and the results are merged into a single risk grade.

Models such as DistilBERT, BiLSTM, DeepFace CNN, Wav2Vec2 and XGBoost were integrated to make the analysis more accurate and the support more relevant to the person. Users get conversational help and recommendations, and those at high risk can reach a professional through a secure session. Making support easy to reach and monitoring continuous should let people get help before a bad day becomes a crisis. The work also shows that multimodal AI is practical for mental wellness monitoring in a realistic setting, although the current version is a prototype and still needs larger trials with real users.

8.2 Future Scope

The accuracy of emotion prediction and the quality of conversation could improve with newer deep learning and generative models. Support in more languages, voice-based interaction, stronger facial emotion recognition and personalised therapy suggestions are natural next steps.

Further work could add IoT wearables for continuous monitoring, a dedicated mobile app, cloud deployment that scales, and real-time emergency alerts. Explainable AI, bias detection and models that adapt to each user would also make the system more transparent and easier to trust, moving Healthly closer to a dependable real-time mental healthcare support service.

REFERENCES

[1] Yiyao Zhang, Jiayu Huang, and Ziyi Liu, "AI-Edu-App: An Integrated AI Platform for K-12 Learning Enhancement and Mental Health Support," International Journal of Artificial Intelligence in Education, vol. 12, no. 3, pp. 45–58, 2025.

[2] Naresh Soni, Kaifa Sabana, Yuvraj, Nikhil Lenka, Rohit Kumar, and Amol Purohit, "Revolutionizing Mental Wellness: Buddy – An AI Powered Mental Health Chatbot," International Journal of Advanced Research in Computer Science and Software Engineering, vol. 15, no. 2, pp. 20–28, 2025.

[3] Sumaiya Yeasmin, Mst Masuma Akter Semi, Moustaq Karim Khan Rony, Srabani Das, Anseena Anees Sabeena, Rukshanda Rahman, Barna Biswas, Fahad Ahmed, and Adib Hossain, "Artificial Intelligence for Mental Health Monitoring: A Solution for Digital Behavioral Health Care and Education," Journal of Healthcare Informatics Research, vol. 8, no. 1, pp. 60–74, 2026.

[4] Sumaiya Yeasmin, Srabani Das, Md Fakhrul Hasan Bhuiyan, Samia Hasan Suha, Mani Prabha, Nur Vanu, and Arif Hosen, "Artificial Intelligence in Mental Health: Leveraging Machine Learning for Diagnosis, Therapy, and Emotional Well-being," International Journal of Machine Learning Applications, vol. 10, no. 4, pp. 120–132, 2025.

[5] Xiaofa Zheng, FuMo Luo, Jie Chang, Yunpeng Jia, and Jun Chen, "Design and Implementation of an Intelligent Emotional Analysis and Healing System for University Students," International Journal of Emotional Computing and Human Interaction, vol. 7, no. 2, pp. 33–46, 2025.

[6] "Enhancing Mental Health with Artificial Intelligence: Current Trends and Future Prospects," Journal of Artificial Intelligence and Healthcare Systems, vol. 9, no. 1, pp. 15–27, 2024.

[7] Eliane M. Boucher, Nicole R. Harake, Haley E. Ward, Sarah Elizabeth Stoeckl, Junielly Vargas, Jared Minkel, Acacia C. Parks, and Ran Zilca, "Artificially Intelligent Chatbots in Digital Mental Health Interventions," Nature Digital Medicine, vol. 4, no. 15, pp. 1–9, 2021.

[8] Vijayalakshmi S., Disney Sandhya A., and Nimala Deve P., "AI-Powered Holistic Mental Health Monitoring Using Facial Emotion Recognition and Voicebot for Personalized Support," in

Proceedings of the International Conference on Smart Healthcare Technologies, pp. 101–110, 2025.

[9] Ayodeji Olusegun Ibitoye, Shaiyini Ravindran, Oluwaseyi Funmi Afe, and Adeyinka O. Abiodun, "Predicting Digital Addiction Patterns with Machine Learning for Personalized Mental Health Support," in Proceedings of the IEEE International Conference on Data Science and Artificial Intelligence, pp. 220–228, 2025.

[10] Yongsheng Ma, Xianhui Sun, and Aiqun Ma, "PSG-Pair: A Psychological Snapshot Guided Model for Wellness Communities," in Proceedings of the International Conference on Natural Language Processing and AI Systems, pp. 90–102, 2025.

[11] Rawan AlMakinah, Andrea Norcini-Pala, Lindsey Disney, and M. Abdullah Canbaz, "Enhancing Mental Health Support through Human-AI Collaboration: Toward Secure and Empathetic AI-Enabled Chatbots," arXiv:2410.02783, 2024.

[12] Xu Zhang, Xiangcheng Zhang, Weisi Chen, Chenlong Li, and Chengyuan Yu, "Improving Speech Depression Detection Using Transfer Learning with wav2vec 2.0 in Low-Resource Environments," Scientific Reports, vol. 14, article 9543, 2024, doi: 10.1038/s41598-024-60278-1.

[13] Thien Vu, Research Dawadi, Masaki Yamamoto, Jie Ting Tay, Naoki Watanabe, Yuki Kuriya, Ai Oya, Phap Ngoc Hoang Tran, and Michihiro Araki, "Prediction of Depressive Disorder Using Machine Learning Approaches: Findings from the NHANES," BMC Medical Informatics and Decision Making, vol. 25, article 83, 2025, doi: 10.1186/s12911-025-02903-1.

[14] Jaedong Oh, Mirae Kim, Hyejin Park, and Hayoung Oh, "Are You Depressed? Analyze User Utterances to Detect Depressive Emotions Using DistilBERT," Applied Sciences, vol. 13, no. 10, article 6223, 2023, doi: 10.3390/app13106223.

[15] Anuja Pinge, Vinaya Gad, Dheryta Jaisighani, Surjya Ghosh, and Sougata Sen, "Detection and Monitoring of Stress Using Wearables: A Systematic Review," Frontiers in Computer Science, vol. 6, article 1478851, 2024, doi: 10.3389/fcomp.2024.1478851.

[16] Nuno Gomes, Matilde Pato, André Ribeiro Lourenço, and Nuno Datia, "A Survey on Wearable Sensors for Mental Health Monitoring," Sensors, vol. 23, no. 3, article 1330, 2023, doi: 10.3390/s23031330.

[17] M. Srisankar and K. P. Lochanambal, "Deep Learning Models of LSTM-ANN and BiLSTM-ANN for Classification Accuracy," International Journal of Computer Applications, vol. 186, no. 3, pp. 22–27, 2024.

[18] Thomas Sounack, Joshua Davis, Brigitte Durieux, Antoine Chaffin, Tom J. Pollard, Eric Lehman, Alistair E. W. Johnson, Matthew McDermott, Tristan Naumann, and Charlotta Lindvall, "BioClinical ModernBERT: A State-of-the-Art Long-Context Encoder for Biomedical and Clinical NLP," arXiv:2506.10896, 2025.

[19] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf, "DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter," in Proceedings of the 5th Workshop on Energy Efficient Machine Learning and Cognitive Computing (EMC2), 2019, arXiv:1910.01108.

[20] Tianqi Chen and Carlos Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794, 2016, doi: 10.1145/2939672.2939785.

APPENDIX

i) Plagiarism Report

ii) Conference/Journal Paper and Certificate

iii) Project Exhibition Certificate